

The exact relation-window spectrum and its arithmetic determinant bound

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This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

1 The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi-\hat{\psi})} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, & c_a &= 2^{1-2a}\Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi \mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q - 1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned}\mathcal{B}(t) &= \log \frac{V_{q-1}(t)V_q(t)}{V_{2q-1}(t)V_{2q}(t)}, \\ C(t) &= M(t)^{-1}(M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q).\end{aligned}\tag{AW5}$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta\mathcal{B} = \mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt.\tag{AW6}$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

2 Full spectra and an explicit source isometry

Theorem 1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q - 1.\tag{AW7}$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi\mathcal{P}_0 \subset F_1 = \chi\mathcal{P}_{q-1} \subset F_2 = \chi\mathcal{P}_q \subset H.\tag{AW8}$$

Their dimensions are $1, q, q + 1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by $1, 2, 1, 0$. The dimensions are respectively $1, q - 1, 1, q$.

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by $0, 1, 2, 1$, with dimensions $q, 1, q - 1, 1$. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j.\tag{AW9}$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q} R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} = r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \tag{AW10}$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\mathrm{Tr}((U - W)C) = \mathrm{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \tag{AW11}$$

Indeed cyclicity moves T in $\mathrm{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \tag{AW12}$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\mathrm{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

3 The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \tag{AW13}$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Theorem 2. *On the unchanged arithmetic source,*

$$|\Delta \mathcal{B}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \tag{AW14}$$

In particular the coefficient $2q$ in AT18 can be improved to $2q - 1$, while the first bound retains the full relative spectrum.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, Pe_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|Pe_j\|^2$ and $\mathrm{Tr} P = r$. Thus $\mathrm{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AW15})$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{AW16})$$

No mutual commutation of U, W, C is used.

For the actual interpolated metric, $C(t) = ((1-t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + tb_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + tb_j)$ and has integral $\log b_j$. Integrating (AW16) in the exact signed identity (AW6) proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q - 1) = 2q - 1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1 - t + tc)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^*MF$, all subspaces have their F^{-1} images, and $D' = F^{-1}DF$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m - 1)](k + 1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \mathcal{L}_q(D). \quad (\text{AW18})$$

This is substitution of the complete supplied Gamma theorem into the proved first inequality of (AW14). It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

4 The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi\mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}}e$, since $J_2e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \quad (\text{AW19})$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$, contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_{\mathcal{N}})|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor and integrate (AW17). The exact nonlinear endpoint bound is

$$\left| 2 \operatorname{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/2}) \right| \leq \log(b_3/b_1). \quad (\text{AW22})$$

Equivalently, with the original values $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$2 \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2 \log \cosh(A_0 + L). \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

5 Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.